

DEEPU DINESH

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PROFESSIONAL SUMMARY

Enthusiastic BSc (Hons) Software Development student at Technological University of Shannon (Moylish Campus) with a strong foundation in programming (C#, C++, Java, Python) and web/app development. Skilled in creating responsive, user-centric applications using HTML, CSS, Bootstrap, PHP, and JavaScript. Experienced with 3D graphics and game engines, including Unity3D, Unreal Engine and Blender. Eager to apply technical and problem-solving skills to real-world projects and innovative software solutions.

SKILLS

C, C++, C#, Java, Python, Docker, WSL2, Bash, HTML, CSS, Bootstrap, JavaScript, PHP, MySQL, Unity, Unreal, Blender, Photoshop

WORK EXPERIENCE

Jaguar Land-Rover, Shannon

Jan 2026 – Aug 2026

Software Development Intern

- Developed software within Linux-based development environments using Python, C and C++, often using TMUX, Docker and command-line tooling using Bash.
- Investigated and **enabled execution of an undocumented application** collaborating extensively with colleagues in software testing departments to resolve blockers and producing documentation to make the setup reproducible for future users.
- Independently designed and developed a **Signal Gateway** written in Python and contributed to its receiver test application written in C, integrating BLF/DBC-based CAN signal processing with a real-time signal management library to be deployed/tested across future vehicular platforms.
- Worked with teammates to investigate unfamiliar systems, communicate technical blockers and integrate independently developed components into the wider software workflow.

Freelance Developer, Fiverr

2022 – 2023

Profile Link: www.fiverr.com/tyrobyte

Gig Reviews Link: www.fiverr.com/tyrobyte/create-c-sharp-scripts-and-simple-features-for-your-unity-projects | www.fiverr.com/tyrobyte/bug-fixing-in-unity

- Designed and developed custom graphical or gameplay focused projects according to client specifications using Unity3D, Blender, Photoshop and a wide variety of tools to adhere to graphical, audio and visual requirements from clients.
- **Delivered 85%** of client orders on schedule and secured a **100% satisfaction rate**, demonstrating efficiency and reliability in project execution.
- Engaged with a diverse international clientele, interpreting requirements through the language barrier while delivering consistent high-quality results, maintaining an excellent satisfaction rating.
- Authored comprehensive visual documentation and practiced effective coding standards, enhancing code readability and expandability.

EDUCATION

BSc (Hons) in Software Development

2023 – Present

Year-1 GPA: 75 | Year-2 GPA: 64

Technological University of Shannon

CERTIFICATES

- Python Specialization Course – University of Michigan (Coursera) **Nov 25, 2020**

PROJECTS

Signal Gateway (JLR)

2026 | Python, C, C++

- Independently developed a **Signal Gateway** in **Python** that replays CAN traffic from a BLF file with realistic timestamp-based pacing, decoding selected channels against their associated DBC files using **python-cantools** using a **multi-threaded** approach.
- Built a **thread-safe** shared data pipeline for raw CAN payloads and manually decoded signals, assembling an IPC PDU payload every 10ms on a dedicated thread.
- Implemented a one time, lookup generation process during application startup stored in memory, mapping signals to CAN message positions, signal groups, IPC PDU offsets and lengths to **optimize the payload generation process**.
- **Developed an algorithm** that returns the bitwise operations to extract individual signals from raw byte arrays using signal group definitions, start bit positions and signal lengths and repurposed it within the Signal Gateway to perform manual packing of certain signals into IPC PDU payloads.
- **Collaborated effectively** with the lead developer to integrate the signal extraction approach into the wider autogenerated signal management workflow, developing effective **logging/debugging strategies** in the process

CLIPlayer

GitHub Repo: <https://github.com/DeepuDinesh05/CLIPlayer>

2026 | Python, PowerShell, Terminal UI

- Developed a terminal-based audio player with modular components for playback control, file loading, metadata extraction, keyboard input and terminal rendering.
- Implemented a cover art display solution for Windows Terminal that uses **mutagen** for audio metadata extraction, **Pillow** for image processing and **Sixel** for terminal display.
- Implemented playlist handling, playback state tracking, seeking and volume control alongside audio metadata extraction.
- Created an ASCII based waveform visualization and PowerShell installation scripts that allow the user to execute the application from the terminal via an alias as a command.

ThreeJS Model Viewer

GitHub Repo: <https://github.com/DeepuDinesh05/ThreeJSModelViewer>

2026 | ThreeJS, HTML, JS, CSS, Bootstrap

- Developed a **3D Model Viewer Web-Application** hosted on GitHub Pages, built for game development usage and learning ThreeJS.
- Built using **HTML, JS, ThreeJS, CSS, Bootstrap** for a responsive and interactive user experience.
- Implemented support for **importing and viewing multiple 3D formats** including **FBX, OBJ, STL, GLTF/GLB**.
- Implemented **export functionality** for supported file formats alongside a **real-time polygon count display** to provide useful insight into model complexity.

HOBBIES

- Strength training & Martial Arts (Judo, Boxing, Calisthenics)
- Sports (Cycling, Badminton)